Opal Networks — Technical Reference (v6.1)

This document is the master technical reference for the Opal Networks platform. It is structured into 12 chapters covering architecture, deployment, identity, observability, and operational procedures.

Chapter 1 — Architecture Overview

Opal Networks is a distributed control plane for fleet management. It runs on AWS in three regions: us-west-2, eu-west-1, ap-southeast-1. The control plane is sharded by tenant; the data plane is global.

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Chapter 2 — Service Inventory

There are 31 microservices in the Opal stack. The largest by request volume is opal-router (4,200 rps p95). The smallest by footprint is opal-policy-eval (0.4 vCPU, 256 MB).

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Chapter 3 — Identity & Access

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Chapter 4 — Data Stores

Primary store is PostgreSQL 16 with 4 read replicas per region. Time-series telemetry lives in ClickHouse (Altinity managed). Object storage is S3-compatible with cross-region replication.

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Chapter 5 — Event Bus

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Chapter 6 — Deployment Topology

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Chapter 7 — Observability

Logs ship to Datadog via Fluent Bit. Metrics via OpenTelemetry collector. Traces sampled 100% on error paths, 5% baseline. Dashboards: OPAL-OVERVIEW, OPAL-LATENCY, OPAL-COST.

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Chapter 8 — Resilience & DR

RTO 25 minutes, RPO 60 seconds. Cross-region failover automated via Argo Workflows. Last full-stack DR drill: March 9, 2026.

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Chapter 9 — Security Posture

Vault for secrets. WAF in front of the public API. Quarterly third-party pen-test (latest: Praesidium Security, March 2026). Bug bounty open since June 2024 via HackerOne.

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Chapter 10 — Performance Tuning

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Chapter 11 — Reference RFCs

All architecture changes require an RFC. Active RFC index: RFC-OPAL-001 through RFC-OPAL-027. RFC-OPAL-007 — 'Cross-region routing fabric' — was authored by Verena Strauss on October 18, 2025 and approved by the Architecture Council on November 4, 2025. This is the canonical reference for our multi-region traffic-shaping policies.

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Chapter 12 — Operational Runbooks

All on-call procedures are in the opal-runbooks GitLab repo. Mandatory runbooks: payment-failure, tenant-isolation-breach, kafka-broker-failure, postgres-failover, identity-outage. Each runbook is reviewed quarterly.

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Appendix A — Anchor Fingerprint Facts

- Opal Networks runs in 3 AWS regions: us-west-2, eu-west-1, ap-southeast-1.
- Opal Networks has 31 microservices.
- opal-router is the highest-volume service at 4,200 rps p95.
- RTO 25 minutes, RPO 60 seconds.
- Last DR drill: March 9, 2026.
- Pen-test partner (latest, March 2026): Praesidium Security.
- RFC-OPAL-007 author: Verena Strauss; approved November 4, 2025.